# Login Security & Attendance Management System

## Python Project

A simple system to protect and manage employee attendance records

Presented by: Vaibhav Rajendra Salunkhe

Course : Cyber Security & Ethical Hacking with AI

# Introduction

## This project is made using Python.

- Creating the Project in VS Code
- It allows only a user with the correct login details to open the system.
- After login, the user can mark attendance and view saved records.
- Attendance is stored in a file named report.log.

# Problem Statement

## Manual attendance can take time.

- Paper records can be lost or damaged.
- Finding old attendance details can be difficult.
- A simple digital record makes attendance easier to manage.

# Project Objectives

- Create a simple username and password check.
- Allow the user to mark employee attendance.
- Save employee details in a file.
- Display all saved attendance records.
- Practise Python conditions, loops and file handling.

# Tools and Technologies

## Programming language: Python 3

- Editor: IDLE / VS Code
- Storage: Text file (report.log)
- Python concepts: input, variables, if-else, while loop, break and file handling

# System Requirements

## A computer or laptop

Python 3 installed

A code editor (VS Code) or Python IDLE

Basic knowledge of running a Python program

# How the System Works

1. The program asks for a username and password.
2. It checks the entered details.
3. If login is correct, the attendance menu opens.
4. The user can mark attendance or view records.
5. The user can exit the system.

# Login Security

- Correct username: admin
- Correct password: root
- Correct details: Login Successful
- Wrong password: Incorrect Password
- Wrong username: User not found

# Attendance Details

- Employee name
- Attendance status: Present or Absent
- Department
- Date
- In-time
- Out-time

# Menu Options

Option 1 — Mark Attendance: enter employee details and save them.

Option 2 — View Attendance: read and display saved records.

Option 3 — Exit: close the menu using break.

# File Handling

- File name: report.log
- Write mode (w): creates or replaces the file content.
- Append mode (a): adds a new record at the end.
- Read mode (r): reads and displays the records.
- Each attendance record is stored on a new line.

# Sample Attendance Record

Vaibhav - P - Admin - 28/03/26 - 10:00 - 18:00

Rahul - Ab - HR - 28/03/26 - 10:00 - 18:00

Format: Name - Status - Department - Date - In-time - Out-time

P = Present, Ab = Absent

# Python Concepts Used

- Variables store login and attendance details.
- input() takes information from the user.
- if-else checks login and menu choices.
- while True repeats the menu.
- break stops the loop.
- open() and write/read methods handle the file.

# Testing and Limitations

Test 1: correct login opens the menu.

Test 2: wrong login does not open the menu.

Test 3: Mark Attendance adds a record.

Test 4: View Attendance displays file contents.

# Conclusion and Future Scope

- The project demonstrates login security and attendance management.
- It saves and displays attendance records using a log file.
- **Future improvements**: preserve old records, validate Present/Absent, add automatic date and time, search employee records, and use a database.